LEARNING HUB MASTER SERIES • LINEAR ALGEBRA • VOLUME 07

Symmetric Matrices & The Spectral Theorem

Universal Orthogonal Diagonalization, Column-Row Spectral Projections, Sylvester's Law of Inertia, and Complex Hermitian Spaces

---

Core Foundations Grounded in:

• Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 6.4)

• Kenji Hiranabe — *The Art of Linear Algebra* (Spectral Column-Row Architecture)

Curriculum Topics: Real Eigenvalue Proof, Orthogonal Eigenvector Theorem, Spectral Factorization $S = Q\Lambda Q^{p}$, Rank-1 Projection Decompositions, Quadratic Inertia Invariance, Spectrum Slicing, Unitary Factorization $U\Lambda U^{d}$.

Authoring: Learning Hub Editorial Board • **Standards:** ISO Paged Media Compliant • **Ed:** Master Reference (EN)

Table of Contents — Volume 07

Part I: Fundamentals of Real Symmetric Matrices

Chapter 1: Symmetry Definition, Inner Products & Physical Motivation
Chapter 2: The Real Eigenvalues Theorem (Formal Proof via Complex Conjugate Transpose)

Part II: Orthogonal Eigenvectors & Rigid Rotations

Chapter 3: The Orthogonal Eigenvectors Theorem (Formal Proof for Distinct Eigenvalues)
Chapter 4: Orthogonal Matrices, Isometric Rigid Motions & Normalization

Part III: The Spectral Theorem & Universal Diagonalizability

Chapter 5: The Spectral Theorem ($S = Q\Lambda Q^p$, Inductive Proof for Repeated Eigenvalues)
Chapter 6: The 3-Stage Geometric Pipeline (Rotation Q^p → Stretch Λ → Rotation Q) [Figure 7.1]

Part IV: Spectral Projection Decomposition & Functional Calculus

Chapter 7: Rank-1 Projection Decomposition ($S = \sum \lambda_i q_i q_i^p$, The 5 Algebraic Laws) [Figure 7.2]
Chapter 8: Spectral Functional Calculus (Decoupled Matrix Powers, Inverses & Norms)

Part V: Sylvester's Law of Inertia & Complex Extensions

Chapter 9: Congruence Transformations, Pivot Signs vs. Eigenvalue Signs & Homotopy [Figure 7.3]
Chapter 10: Spectrum Slicing (Bisection) & Complex Hermitian Matrices ($H = U\Lambda U^d$) [Figure 7.4]

Part VI: Fast-Track Hand Calculation Shortcuts ⚡ 7 Golden Fast-Tracks

⚡ Fast-Track 7.1: Instant 2×2 Symmetric Eigenvalues via Center & Radius Formula
⚡ Fast-Track 7.2: Rapid Orthonormal Basis for Repeated Eigenvalues in $\mathbb{R}^3$ via Cross Product
⚡ Fast-Track 7.3: Instant Rank-1 Spectral Projection Calculation & Complement Rule
⚡ Fast-Track 7.4: 10-Second Sylvester Inertia Counting without Characteristic Polynomials
⚡ Fast-Track 7.5: Spectral 2-Norm and Frobenius Norm Direct Extraction from Eigenvalues
⚡ Fast-Track 7.6: Commuting Symmetric Matrices & Simultaneous Diagonalization
⚡ Fast-Track 7.7: Spectral Inversion & Determinant Reconstruction via Projection Pieces

Part VII: Rigorous Step-by-Step Worked Problems

Problem 7.1: Complete Orthogonal Diagonalization of a 2×2 Symmetric Matrix
Problem 7.2: Rank-1 Spectral Projection Decomposition & Functional Calculus Powers
Problem 7.3: 3×3 Symmetric Matrix with Repeated Eigenvalue (Geometric Multiplicity Proof)
Problem 7.4: Sylvester's Inertia Verification & Spectrum Slicing Bisection Method
Problem 7.5: Complex Hermitian Diagonalization via Unitary Transformation Matrix

Part VIII: Comprehensive Master Reference Matrix

Symmetric vs. General Matrix Structural Comparison Table & Final Exam Checklist

Part I: Fundamentals of Real Symmetric Matrices

Chapter 1: Symmetry Definition, Inner Products & Physical Motivation

In general linear algebra, square matrices can possess complex eigenvalues, skewed eigenvectors that fail to meet at right angles, and severe degeneracies where algebraic multiplicity exceeds geometric multiplicity ("defective" matrices requiring Jordan blocks). However, the moment a matrix is **real and symmetric**, an extraordinary structural harmony emerges.

Definition 7.1: Real Symmetric Matrix

A square matrix $\mathbf{S} \in \mathbb{R}^{n \times n}$ is defined to be **symmetric** if and only if it equals its own transpose:

$$\mathbf{S}^{\mathsf{T}} = \mathbf{S} \quad \Leftrightarrow \quad s_{ij} = s_{ji} \quad \text{for all} \quad 1 \le i, j \le n.$$

The off-diagonal entries across the main diagonal are mirror images of each other. The space of all real symmetric $n \times n$ matrices forms a vector subspace of dimension $\frac{n(n+1)}{2}$.

Physical & Computational Ubiquity: Real symmetric matrices arise universally in science and engineering:

- **Structural Mechanics & Physics:** Stiffness matrices $\mathbf{K}$ in elasticity, inertia tensors $\mathbf{I}$ in rigid-body dynamics, and conductivity tensors are naturally symmetric due to Maxwell-Betti reciprocity.
- **Statistics & Data Science:** Sample covariance matrices $\mathbf{C} = \frac{1}{N-1}\mathbf{X}^{\mathsf{T}}\mathbf{X}$ and correlation matrices are inherently symmetric.
- **Network Theory:** Adjacency matrices of undirected graphs and graph Laplacians $\mathbf{L} = \mathbf{D} - \mathbf{A}$ are symmetric.
- **Optimization & Calculus:** The Hessian matrix $\mathbf{H}$ of second partial derivatives $H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ is symmetric by Clairaut's theorem on mixed partials.

The Fundamental Adjoint Property

For any real matrix $\mathbf{A}$, the standard Euclidean dot product satisfies $\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \mathbf{x}^{\mathsf{T}}(\mathbf{A}\mathbf{y}) = (\mathbf{A}^{\mathsf{T}}\mathbf{x})^{\mathsf{T}}\mathbf{y} = \langle \mathbf{A}^{\mathsf{T}}\mathbf{x}, \mathbf{y} \rangle$. When $\mathbf{S}$ is symmetric, $\mathbf{S}^{\mathsf{T}} = \mathbf{S}$, leading to the profound identity:

$$\mathbf{x}^{\mathsf{T}}\mathbf{S}\mathbf{y} = (\mathbf{S}\mathbf{x})^{\mathsf{T}}\mathbf{y} \quad \text{for all} \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

This self-adjointness means the matrix operator can shift across the inner product between the left and right vectors with zero modification. Every fundamental property of symmetric matrices flows directly from this single algebraic equation.

Chapter 2: The Real Eigenvalues Theorem

Non-symmetric real matrices frequently have complex eigenvalues. For example, a 90° rotation matrix $\mathbf{R} = \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix}$ has characteristic polynomial $\lambda^2 + 1 = 0$, giving pure imaginary eigenvalues $\pm i$ because no real vector can maintain its direction after a quarter-turn. In sharp contrast, a symmetric matrix can *never* rotate space in this circulating manner.

Theorem 7.1: Reality of Eigenvalues

Every eigenvalue λ of a real symmetric matrix $\mathbf{S}$ is strictly real:

$$\mathbf{S} \in \mathbb{R}^{n \times n}, \quad \mathbf{S}^{\mathsf{T}} = \mathbf{S} \quad \Rightarrow \quad \lambda \in \mathbb{R} \quad (\mathrm{Im}(\lambda) = 0).$$

Consequently, all roots of the characteristic polynomial $\det(\mathbf{S} - \lambda\mathbf{I}) = 0$ are real numbers, and all eigenvectors can be chosen with completely real components.

Rigorous Formal Proof (via Complex Conjugate Transpose):

Let $\lambda \in \mathbb{C}$ be an eigenvalue of $\mathbf{S}$, and let $\mathbf{x} \in \mathbb{C}^n$ be its corresponding non-zero eigenvector, so that:

$$\mathbf{S}\,\mathbf{x} = \lambda\,\mathbf{x} \quad (\text{with } \mathbf{x} \neq \mathbf{0}). \qquad [\text{Eq. 1}]$$

Take the complex conjugate of both sides of Eq. 1. Since $\mathbf{S}$ has purely real entries, $\mathbf{S} = \mathbf{S}$:

$$\mathbf{S}\,\mathbf{x} = \lambda\,\mathbf{x}.$$

Now transpose this conjugated equation. Using the rule $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\,\mathbf{A}^{\mathsf{T}}$ and the symmetry $\mathbf{S}^{\mathsf{T}} = \mathbf{S}$:

$$(\mathbf{S}\,\mathbf{x})^{\mathsf{T}} = \mathbf{x}^{\mathsf{T}}\,\mathbf{S}^{\mathsf{T}} = \mathbf{x}^{\mathsf{T}}\,\mathbf{S} = \lambda\,\mathbf{x}^{\mathsf{T}}. \qquad [\text{Eq. 2}]$$

Multiply Eq. 2 on the right by the original eigenvector $\mathbf{x}$:

$$\mathbf{x}^{\mathsf{T}}\,\mathbf{S}\,\mathbf{x} = \lambda\,(\mathbf{x}^{\mathsf{T}}\,\mathbf{x}). \qquad [\text{Eq. 3}]$$

Meanwhile, multiply Eq. 1 on the left by the conjugate transpose row vector $\mathbf{x}^{\mathsf{T}}$ (often denoted $\mathbf{x}^{\mathsf{H}}$):

$$\mathbf{x}^{\mathsf{T}}\,\mathbf{S}\,\mathbf{x} = \lambda\,(\mathbf{x}^{\mathsf{T}}\,\mathbf{x}). \qquad [\text{Eq. 4}]$$

Subtract Eq. 3 from Eq. 4:

$$(\lambda - \lambda)\,(\mathbf{x}^{\mathsf{T}}\,\mathbf{x}) = 0.$$

We now examine the scalar quantity $\mathbf{x}^{\mathsf{T}}\,\mathbf{x} = \mathbf{x}^{\mathsf{H}}\,\mathbf{x}$. If $\mathbf{x}$ has complex components $[x_1, \ldots, x_n]^{\mathsf{T}}$, then:

$$\mathbf{x}^{\mathsf{T}}\,\mathbf{x} = |x_1|^2 + |x_2|^2 + \ldots + |x_n|^2 = \sum_{j=1}^{n} |x_j|^2.$$

Since $\mathbf{x}$ is a non-zero eigenvector, at least one component is non-zero, guaranteeing $\mathbf{x}^{\mathsf{T}}\,\mathbf{x} > 0$ strictly. Therefore, the scalar factor $(\lambda - \lambda)$ must vanish identically:

$$\lambda - \lambda = 0 \quad \Rightarrow \quad \lambda = \lambda.$$

A complex number that equals its own complex conjugate is purely real. Hence $\lambda \in \mathbb{R}$. Since $(\mathbf{S} - \lambda\mathbf{I})$ is now a real matrix with $\det(\mathbf{S} - \lambda\mathbf{I}) = 0$, the homogeneous system $(\mathbf{S} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ has non-trivial real solutions. Thus the eigenvectors $\mathbf{x}$ can always be chosen in $\mathbb{R}^n$. ■

Examination Pitfall: Complex Symmetry vs. Hermitian

Do **not** confuse a complex symmetric matrix ($A = A^{\mathsf{T}}$ with complex entries) with a real symmetric matrix or Hermitian matrix! A complex symmetric matrix can have non-real eigenvalues and can even be defective. For example, $\mathbf{A} = \begin{vmatrix} 1 & i \\ i & -1 \end{vmatrix}$ satisfies $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$, but $\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2$, yielding $\lambda = 0$ with only a 1D eigenspace (defective!). Theorem 7.1 strictly requires $\mathbf{S}$ to be **real**.

Part II: Orthogonal Eigenvectors & Rigid Rotations

Chapter 3: The Orthogonal Eigenvectors Theorem

For a general non-symmetric matrix $\mathbf{A}$, eigenvectors corresponding to distinct eigenvalues are linearly independent, but they are rarely perpendicular. For real symmetric matrices, linear independence is elevated to **mutual orthogonality**.

Theorem 7.2: Orthogonality of Eigenvectors

Let $\mathbf{S}$ be a real symmetric matrix. If $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors corresponding to two *distinct* eigenvalues $\lambda_1 \neq \lambda_2$, then $\mathbf{x}_1$ and $\mathbf{x}_2$ are orthogonal:

$$\lambda_1 \neq \lambda_2 \quad \Rightarrow \quad \mathbf{x}_1^{\mathsf{T}} \mathbf{x}_2 = 0 \quad (\mathbf{x}_1 \perp \mathbf{x}_2).$$

Rigorous Formal Proof:

Suppose $\mathbf{S}\,\mathbf{x}_1 = \lambda_1\,\mathbf{x}_1$ and $\mathbf{S}\,\mathbf{x}_2 = \lambda_2\,\mathbf{x}_2$, with $\lambda_1 \neq \lambda_2$ and both vectors in $\mathbb{R}^n$. Evaluate the triple matrix-vector product $\mathbf{x}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{x}_2$ in two different associative groupings:

Grouping 1: Apply $\mathbf{S}$ to $\mathbf{x}_2$ first:

$$\mathbf{x}_1^{\mathsf{T}} (\mathbf{S}\,\mathbf{x}_2) = \mathbf{x}_1^{\mathsf{T}} (\lambda_2\,\mathbf{x}_2) = \lambda_2\,(\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2).$$

Grouping 2: Apply $\mathbf{S}$ to $\mathbf{x}_1$ using transpose symmetry $\mathbf{S} = \mathbf{S}^{\mathsf{T}}$:

$$\mathbf{x}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{x}_2 = (\mathbf{S}^{\mathsf{T}}\,\mathbf{x}_1)^{\mathsf{T}}\,\mathbf{x}_2 = (\mathbf{S}\,\mathbf{x}_1)^{\mathsf{T}}\,\mathbf{x}_2 = (\lambda_1\,\mathbf{x}_1)^{\mathsf{T}}\,\mathbf{x}_2 = \lambda_1\,(\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2).$$

Since the two groupings represent the exact same scalar value, equate them:

$$\lambda_1\,(\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2) = \lambda_2\,(\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2) \quad \Rightarrow \quad (\lambda_1 - \lambda_2)\,(\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2) = 0.$$

By hypothesis, the eigenvalues are distinct, meaning $\lambda_1 - \lambda_2 \neq 0$. Therefore, the inner product must equal zero:

$$\mathbf{x}_1^{\mathsf{T}}\,\mathbf{x}_2 = 0 \quad \Leftrightarrow \quad \mathbf{x}_1 \perp \mathbf{x}_2. \quad \blacksquare$$

Chapter 4: Orthogonal Matrices, Rigid Motions & Normalization

Because distinct eigenvectors are naturally perpendicular, normalizing each eigenvector $\mathbf{q}_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|}$ yields an **orthonormal basis** for $\mathbf{R}^n$. Assembling these vectors as columns forms an **orthogonal matrix** $\mathbf{Q} = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \dots \quad \mathbf{q}_n]$.

Properties of Orthogonal Matrices Q

An $n \times n$ matrix $\mathbf{Q}$ is orthogonal if its columns are mutually orthonormal: $\mathbf{q}_i^{\mathsf{T}}\,\mathbf{q}_j = \delta_{ij}$ (where $\delta_{ij} = 1$ if $i = j$, and 0 if $i \neq j$).

- **Fundamental Inversion Law:** $\mathbf{Q}^{\mathsf{T}}\,\mathbf{Q} = \mathbf{I} \quad \Rightarrow \quad \mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}$. Transposing computes the inverse for free!
- **Row Orthonormality:** Since $\mathbf{Q}\,\mathbf{Q}^{-1} = \mathbf{I}$, we also have $\mathbf{Q}\,\mathbf{Q}^{\mathsf{T}} = \mathbf{I}$. The rows are equally orthonormal.
- **Length & Angle Preservation (Isometry):** For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

  $$\|\mathbf{Q}\mathbf{x}\|^2 = (\mathbf{Q}\mathbf{x})^{\mathsf{T}}\,(\mathbf{Q}\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\,\mathbf{Q}^{\mathsf{T}}\,\mathbf{Q}\,\mathbf{x} = \mathbf{x}^{\mathsf{T}}\,\mathbf{I}\,\mathbf{x} = \|\mathbf{x}\|^2.$$

  Multiplication by $\mathbf{Q}$ is a rigid rotation or reflection: it never stretches lengths or distorts angles.
- **Determinant:** $\det(\mathbf{Q}^{\mathsf{T}}\,\mathbf{Q}) = \det(\mathbf{Q})^2 = \det(\mathbf{I}) = 1 \Rightarrow \det(\mathbf{Q}) = \pm 1$ (+1 for pure rotations, −1 for reflections).

Part III: The Spectral Theorem & Universal Diagonalizability

Chapter 5: The Spectral Theorem

We now arrive at the pinnacle theorem of finite-dimensional linear algebra. While general matrices require distinct eigenvalues or careful checks on eigenspace dimensions to ensure diagonalizability, symmetric matrices are **universally diagonalizable**.

Theorem 7.3: The Spectral Theorem for Real Symmetric Matrices

Every real symmetric matrix $\mathbf{S} \in \mathbb{R}^{n\times n}$ can be factored into:

$$\mathbf{S} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^{\mathsf{T}} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^{-1},$$

where $\mathbf{Q}$ is an orthogonal matrix whose columns are orthonormal eigenvectors of $\mathbf{S}$, and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is a diagonal matrix containing the real eigenvalues.

Universal Multiplicity Equality: For every eigenvalue λ of a real symmetric matrix, the geometric multiplicity *always* equals the algebraic multiplicity:

$$\mathrm{GM}(\lambda) = \dim(\operatorname{null}(\mathbf{S} - \lambda\mathbf{I})) = \mathrm{AM}(\lambda) \quad \text{for all } \lambda.$$

Defective symmetric matrices do not exist. Even when eigenvalues are repeated, a full orthonormal set of n eigenvectors always spans $\mathbb{R}^n$.

Inductive Proof for Repeated Eigenvalues (Strang Section 7.2):

If all n eigenvalues are distinct, Theorem 7.2 immediately guarantees n mutually perpendicular eigenvectors, proving the theorem. We must prove the theorem holds when eigenvalues are repeated. We proceed by mathematical induction on the matrix dimension n.

Base Step (n = 1): Any 1×1 matrix $[s_{11}]$ is trivially diagonal with $Q = [1]$.

Inductive Step: Assume every real symmetric $(n-1)\times(n-1)$ matrix is orthogonally diagonalizable. By the Fundamental Theorem of Algebra and Theorem 7.1, $\mathbf{S}$ has at least one real eigenvalue λ_1 and a corresponding unit eigenvector $\mathbf{q}_1$ ($\|\mathbf{q}_1\| = 1$) such that $\mathbf{S}\,\mathbf{q}_1 = \lambda_1\,\mathbf{q}_1$.

Using Gram-Schmidt, extend $\mathbf{q}_1$ to an orthonormal basis for $\mathbb{R}^n$: $\mathbf{Q}_1 = [\mathbf{q}_1 \quad \mathbf{V}]$, where $\mathbf{V} \in \mathbb{R}^{n\times(n-1)}$ satisfies $\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{Q}_1 = \mathbf{I}$. Compute the similarity transform $\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1$:

$$\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1 = \begin{vmatrix} \mathbf{q}_1^{\mathsf{T}} \\ \mathbf{V}^{\mathsf{T}} \end{vmatrix} \mathbf{S} \begin{vmatrix} \mathbf{q}_1 & \mathbf{V} \end{vmatrix} = \begin{vmatrix} \mathbf{q}_1^{\mathsf{T}}\mathbf{S}\,\mathbf{q}_1 & \mathbf{q}_1^{\mathsf{T}}\mathbf{S}\,\mathbf{V} \\ \mathbf{V}^{\mathsf{T}}\mathbf{S}\,\mathbf{q}_1 & \mathbf{V}^{\mathsf{T}}\mathbf{S}\,\mathbf{V} \end{vmatrix}.$$

Examine each block:

- Top-left: $\mathbf{q}_1^{\mathsf{T}}(\mathbf{S}\,\mathbf{q}_1) = \mathbf{q}_1^{\mathsf{T}}(\lambda_1\,\mathbf{q}_1) = \lambda_1\,\|\mathbf{q}_1\|^2 = \lambda_1$.
- Bottom-left: $\mathbf{V}^{\mathsf{T}}(\mathbf{S}\,\mathbf{q}_1) = \lambda_1(\mathbf{V}^{\mathsf{T}}\,\mathbf{q}_1) = \mathbf{0}$, because columns of $\mathbf{V}$ are orthogonal to $\mathbf{q}_1$.
- Top-right: Transpose of bottom-left because $(\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1)^{\mathsf{T}} = \mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1$ is symmetric! Thus $\mathbf{q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{V} = \mathbf{0}^{\mathsf{T}}$.

Therefore, $\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1$ decouples into block-diagonal form:

$$\mathbf{Q}_1^{\mathsf{T}}\,\mathbf{S}\,\mathbf{Q}_1 = \begin{vmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{S}_{n-1} \end{vmatrix}, \quad \text{where } \mathbf{S}_{n-1} = \mathbf{V}^{\mathsf{T}}\,\mathbf{S}\,\mathbf{V}.$$

Since $\mathbf{S}_{n-1}$ is symmetric and of size $(n-1)\times(n-1)$, our inductive hypothesis applies: there exists an orthogonal matrix $\mathbf{Q}_{n-1}$ such that $\mathbf{Q}_{n-1}^{\mathsf{T}}\,\mathbf{S}_{n-1}\,\mathbf{Q}_{n-1} = \mathbf{\Lambda}_{n-1}$. Defining the full orthogonal matrix:

$$\mathbf{Q} = \mathbf{Q}_1 \begin{vmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{Q}_{n-1} \end{vmatrix} \quad \Rightarrow \quad \mathbf{Q}^T \mathbf{S}\, \mathbf{Q} = \begin{vmatrix} \lambda_1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{\Lambda}_{n-1} \end{vmatrix} = \mathbf{\Lambda}.$$

Thus $\mathbf{S} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^T$ holds for all n. Zero defective symmetric matrices exist. ■

Chapter 6: The 3-Stage Geometric Pipeline

The factorization $\mathbf{S} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^T$ reveals the exact geometric action of any symmetric matrix upon an input vector $\mathbf{x}$. Applying $\mathbf{S}$ to $\mathbf{x}$ decomposes into a strictly uncoupled 3-stage process reading from right to left:

1. **Stage 1: Rigid Coordinate Rotation by $\mathbf{Q}^T$:** The vector $\mathbf{x}$ is rotated into the eigenbasis coordinates $\mathbf{y} = \mathbf{Q}^T\mathbf{x}$. Because $\mathbf{Q}$ is orthogonal, this rotation preserves lengths and angles without distortion.
2. **Stage 2: Independent Axial Stretching by $\mathbf{\Lambda}$:** In eigen-coordinates, the matrix $\mathbf{\Lambda}$ scales each coordinate independently: $y_i \mapsto \lambda_i y_i$. There is zero cross-coupling between coordinates.
3. **Stage 3: Inverse Rigid Coordinate Rotation by $\mathbf{Q}$:** The scaled vector is rotated back to standard coordinates via $\mathbf{Q}$.

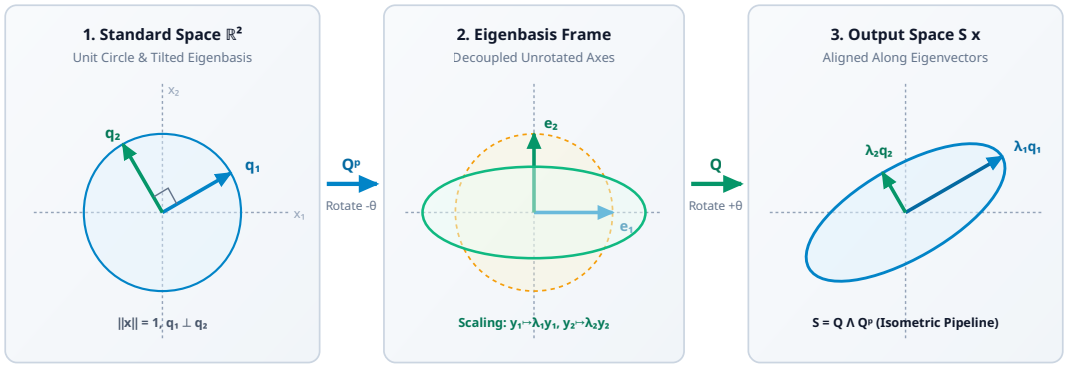


Figure 7.1: *The 3-stage geometric pipeline of orthogonal diagonalization $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$. A unit sphere is rotated by $\mathbf{Q}^T$ into the eigenframe, stretched along decoupled axes by eigenvalues λ_1, λ_2, and rotated back by $\mathbf{Q}$ to form an orthogonal ellipse.*

Part IV: Spectral Projection Decomposition & Functional Calculus

Chapter 7: Rank-1 Projection Decomposition

In *The Art of Linear Algebra*, Kenji Hiranabe highlights that matrix multiplication $\mathbf{AB}$ can be interpreted as the sum of column-by-row outer products. Expanding the Spectral Theorem $\mathbf{S} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^T$ through this column-row lens decomposes the entire matrix into a weighted sum of rank-1 projection operators.

Theorem 7.4: Spectral Decomposition into Projection Matrices

Let $\mathbf{S} = \mathbf{Q}\,\mathbf{\Lambda}\,\mathbf{Q}^T$, where $\mathbf{Q}$ has orthonormal columns $\mathbf{q}_1, \ldots, \mathbf{q}_n$. Then:

$$\mathbf{S} = \begin{vmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{vmatrix} \begin{vmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{vmatrix} \begin{vmatrix} \mathbf{q}_1^T \\ \vdots \\ \mathbf{q}_n^T \end{vmatrix} = \sum_{i=1}^{n} \lambda_i \mathbf{q}_i \mathbf{q}_i^T = \lambda_1 \mathbf{P}_1 + \lambda_2 \mathbf{P}_2 + \ldots + \lambda_n \mathbf{P}_n,$$

where each $\mathbf{P}_i = \mathbf{q}_i \mathbf{q}_i^T$ is the **orthogonal projection matrix** onto the 1-dimensional subspace spanned by eigenvector $\mathbf{q}_i$.

The 5 Fundamental Algebraic Laws of Spectral Projections

The matrices $\mathbf{P}_i = \mathbf{q}_i \mathbf{q}_i^T$ satisfy five universal structural properties:

1. **Rank-1 Property:** $\text{rank}(\mathbf{P}_i) = 1$, since every column is a scalar multiple of $\mathbf{q}_i$.
2. **Symmetry:** $\mathbf{P}_i^T = (\mathbf{q}_i \mathbf{q}_i^T)^T = (\mathbf{q}_i^T)^T \mathbf{q}_i^T = \mathbf{q}_i \mathbf{q}_i^T = \mathbf{P}_i$.
3. **Idempotence (Projection Law):** Projecting twice is identical to projecting once:
$$\mathbf{P}_i^2 = (\mathbf{q}_i \mathbf{q}_i^T)(\mathbf{q}_i \mathbf{q}_i^T) = \mathbf{q}_i (\mathbf{q}_i^T \mathbf{q}_i) \mathbf{q}_i^T = \mathbf{q}_i (1) \mathbf{q}_i^T = \mathbf{P}_i.$$
4. **Mutual Orthogonality:** The subspaces are perpendicular, so projecting onto one destroys components in all others:
$$\mathbf{P}_i \mathbf{P}_j = \mathbf{q}_i (\mathbf{q}_i^T \mathbf{q}_j) \mathbf{q}_j^T = \mathbf{q}_i (0) \mathbf{q}_j^T = \mathbf{O} \quad \text{for all} \quad i \neq j.$$
5. **Resolution of the Identity (Completeness):** Summing all projections reconstructs the identity operator:
$$\sum_{i=1}^{n} \mathbf{P}_i = \sum_{i=1}^{n} \mathbf{q}_i \mathbf{q}_i^T = \mathbf{Q} \mathbf{Q}^T = \mathbf{I}.$$

Geometric Action on a Vector: For any vector $\mathbf{x} \in \mathbb{R}^n$, applying $\mathbf{S}$ yields: $\mathbf{S}\,\mathbf{x} = (\sum \lambda_i \mathbf{P}_i)\,\mathbf{x} = \sum \lambda_i (\mathbf{q}_i^T \mathbf{x})\,\mathbf{q}_i$. The matrix drops perpendiculars from $\mathbf{x}$ onto each principal axis, scales the projection component by eigenvalue λ_i, and vector-adds the orthogonal pieces.

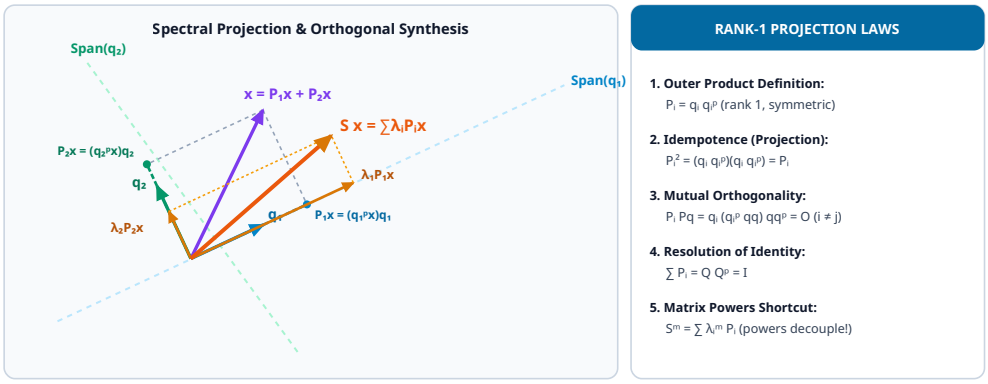


Figure 7.2: *Rank-1 spectral projection decomposition $\mathbf{S} = \sum \lambda_i \mathbf{q}_i \mathbf{q}_i^T$. The vector $\mathbf{x}$ decomposes into perpendicular projection pieces $\mathbf{P}_1\mathbf{x}$ and $\mathbf{P}_2\mathbf{x}$, which are independently scaled by λ_1, λ_2 and summed to produce $\mathbf{Sx}$.*

Chapter 8: Spectral Functional Calculus

Because the spectral projection matrices satisfy $\mathbf{P}_i^2 = \mathbf{P}_i$ and $\mathbf{P}_i \mathbf{P}_j = \mathbf{O}$ for $i \neq j$, computing powers or algebraic functions of a symmetric matrix becomes trivial.

Theorem 7.5: Functional Calculus for Symmetric Matrices

Let $\mathbf{S} = \sum_{i=1}^{n} \lambda_i \mathbf{P}_i$ be the spectral decomposition of $\mathbf{S}$. For any function $f: \mathbb{R} \to \mathbb{R}$ defined on the spectrum of $\mathbf{S}$:

$$f(\mathbf{S}) = \sum_{i=1}^{n} f(\lambda_i) \mathbf{P}_i = \mathbf{Q}\, f(\mathbf{\Lambda})\, \mathbf{Q}^T.$$

The projection matrices $\mathbf{P}_i$ and eigenvectors $\mathbf{q}_i$ remain **completely unchanged**; only the scalar eigenvalues λ_i are transformed by $f(\lambda)$.

Crucial Mathematical Applications:

- **Integer Powers:** $\mathbf{S}^k = \sum_{i=1}^{n} \lambda_i^k \mathbf{P}_i$. Cross-terms vanish: $(\mathbf{P}_1 + \mathbf{P}_2)^2 = \mathbf{P}_1^2 + \mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1 + \mathbf{P}_2^2 = \mathbf{P}_1 + \mathbf{P}_2$.
- **Matrix Inversion:** If all $\lambda_i \neq 0$, then $\mathbf{S}^{-1} = \sum_{i=1}^{n} \frac{1}{\lambda_i} \mathbf{P}_i$.

- **Matrix Square Root:** If all $\lambda_i \geq 0$, the unique positive semidefinite square root is $\mathbf{S}^{1/2} = \sum \sqrt{\lambda_i}\, \mathbf{P}_i$.
- **Matrix Exponential:** $e^{\mathbf{S}t} = \sum_{i=1}^{n} e^{\lambda_i t}\, \mathbf{P}_i$.
- **Spectral 2-Norm:** $\|\mathbf{S}\|_2 = \max_{1 \leq i \leq n} |\lambda_i|$.
- **Frobenius Norm:** $\|\mathbf{S}\|_F^2 = \text{trace}(\mathbf{S}^2) = \sum_{i=1}^{n} \lambda_i^2$.

Part V: Sylvester's Law of Inertia & Complex Extensions

Chapter 9: Congruence Transformations & Sylvester's Law

Computing the exact eigenvalues of a large matrix requires finding the roots of a high-degree polynomial $\det(\mathbf{S} - \lambda \mathbf{I}) = 0$, which is computationally expensive. However, Gaussian elimination $\mathbf{S} = \mathbf{L}\,\mathbf{D}\,\mathbf{L}^{\mathsf{T}}$ requires only $O(n^3)$ arithmetic operations. **Sylvester's Law of Inertia** provides an exact bridge between elimination pivots and eigenvalues.

Definition 7.2: Matrix Congruence & Inertia Triplet

Two symmetric matrices $\mathbf{A}$ and $\mathbf{B}$ are called **congruent** if there exists an invertible matrix $\mathbf{C}$ such that:

$$\mathbf{B} = \mathbf{C}\,\mathbf{A}\,\mathbf{C}^{\mathsf{T}}.$$

The **inertia** of a symmetric matrix $\mathbf{S}$ is the ordered triplet of integers:

$$\text{In}(\mathbf{S}) = (n_+, n_-, n_0),$$

where n_+ is the number of strictly positive eigenvalues, n_- is the number of strictly negative eigenvalues, and n_0 is the multiplicity of zero eigenvalues ($n_+ + n_- + n_0 = n$).

Theorem 7.6: Sylvester's Law of Inertia

Congruence transformations preserve inertia: if $\mathbf{C}$ is invertible, then $\mathbf{C}\,\mathbf{S}\,\mathbf{C}^{\mathsf{T}}$ has the exact same number of positive, negative, and zero eigenvalues as $\mathbf{S}$:

$$\text{In}(\mathbf{C}\,\mathbf{S}\,\mathbf{C}^{\mathsf{T}}) = \text{In}(\mathbf{S}).$$

The Pivot Sign Theorem: In symmetric Gaussian elimination $\mathbf{S} = \mathbf{L}\,\mathbf{D}\,\mathbf{L}^{\mathsf{T}}$ without row exchanges (where $\mathbf{L}$ is unit lower-triangular and $\mathbf{D} = \text{diag}(d_1, \ldots, d_n)$ contains the pivots):

Number of positive pivots in $\mathbf{D}$ = Number of positive eigenvalues of $\mathbf{S}$.

Number of negative pivots in $\mathbf{D}$ = Number of negative eigenvalues of $\mathbf{S}$.

Homotopy Continuity Proof (Gilbert Strang Section 6.4):

Let $\mathbf{S} = \mathbf{L}\,\mathbf{D}\,\mathbf{L}^{\mathsf{T}}$. We construct a continuous matrix family (a homotopy) $\mathbf{S}(t)$ connecting the diagonal pivot matrix $\mathbf{D}$ to $\mathbf{S}$ across parameter $t \in [0, 1]$:

$$\mathbf{L}(t) = t\,\mathbf{L} + (1 - t)\,\mathbf{I}, \quad \text{and} \quad \mathbf{S}(t) = \mathbf{L}(t)\,\mathbf{D}\,\mathbf{L}(t)^{\mathsf{T}}.$$

Notice the boundary endpoints:

- At $t = 0$: $\mathbf{L}(0) = \mathbf{I} \Rightarrow \mathbf{S}(0) = \mathbf{I}\,\mathbf{D}\,\mathbf{I}^{\mathsf{T}} = \mathbf{D}$. Its eigenvalues are the pivots $d_1, \ldots, d_n$.
- At $t = 1$: $\mathbf{L}(1) = \mathbf{L} \Rightarrow \mathbf{S}(1) = \mathbf{L}\,\mathbf{D}\,\mathbf{L}^{\mathsf{T}} = \mathbf{S}$. Its eigenvalues are the true eigenvalues λ_i.

Because $\mathbf{L}$ is unit lower-triangular, $\mathbf{L}(t)$ has 1s on its main diagonal for all $t \in [0, 1]$. Hence $\det(\mathbf{L}(t)) = 1$ identically. By the determinant product rule:

$$\det(\mathbf{S}(t)) = \det(\mathbf{L}(t))\,\det(\mathbf{D})\,\det(\mathbf{L}(t)^{\mathsf{T}}) = (1)\cdot\det(\mathbf{D})\cdot(1) = \det(\mathbf{D}) = d_1\,d_2\,\dots\,d_n.$$

Since $\det(\mathbf{D}) \neq 0$ (assuming non-zero pivots), $\det(\mathbf{S}(t))$ **never vanishes** for any $t \in [0, 1]$.

The eigenvalues $\lambda_i(t)$ are continuous functions of t. In order for an eigenvalue to change sign from positive to negative, by the Intermediate Value Theorem it would have to cross zero ($\lambda = 0$). But if $\lambda_i(t) = 0$ for any t, the product of eigenvalues $\det(\mathbf{S}(t))$ would be 0, which contradicts $\det(\mathbf{S}(t)) = \det(\mathbf{D}) \neq 0$. Therefore, no eigenvalue can ever cross zero! Each eigenvalue maintains its sign strictly from $t = 0$ to $t = 1$. Thus, the signs of the eigenvalues of $\mathbf{S}$ match the signs of the pivots d_i. ■

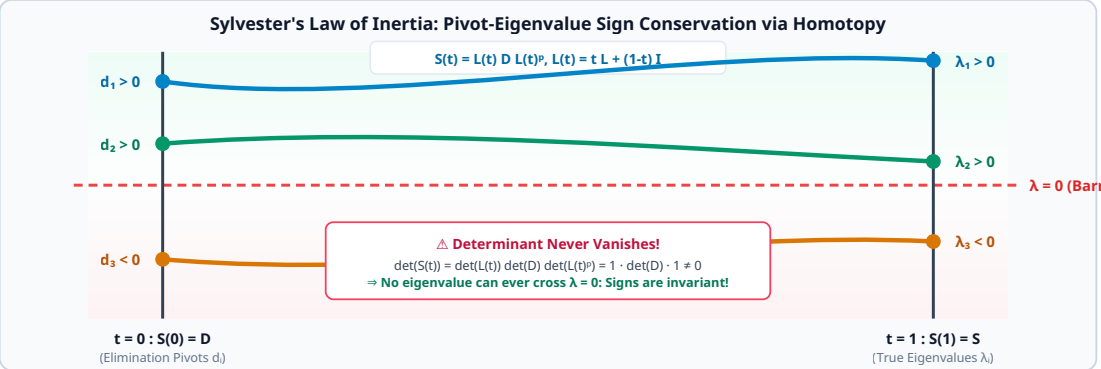


Figure 7.3: *Homotopy proof of Sylvester's Law of Inertia. Because $\det(\mathbf{S}(t)) = \det(\mathbf{D}) \neq 0$, eigenvalue trajectories $\lambda_i(t)$ can never cross the zero barrier $\lambda = 0$, proving conservation of signs between pivots d_i and eigenvalues λ_i.*

Chapter 10: Spectrum Slicing & Complex Hermitian Matrices

The Spectrum Slicing Algorithm (Bisection Method): Sylvester's Law allows counting the exact number of eigenvalues in any arbitrary interval (a, b) in seconds without computing a single root.

⚡ Algorithm: Spectrum Slicing via Shifted Pivots

To find the number of eigenvalues of $\mathbf{S}$ strictly less than a shift threshold $\sigma \in \mathbb{R}$:

1. Form the shifted matrix $\mathbf{S} - \sigma\mathbf{I}$.
2. Perform row elimination without row exchanges to obtain pivots $d_1(\sigma), \dots, d_n(\sigma)$.
3. Count the number of negative pivots:

$$\text{Number of eigenvalues } \lambda < \sigma \quad = \quad \text{Number of negative pivots of } (\mathbf{S} - \sigma\mathbf{I}).$$

To count eigenvalues in an open interval (a, b): $\text{Count}(a < \lambda < b) = \text{NegPivots}(\mathbf{S} - b\mathbf{I}) - \text{NegPivots}(\mathbf{S} - a\mathbf{I})$.

Generalization to Complex Matrices: Hermitian & Unitary Spaces

When elements are complex numbers ($\in \mathbb{C}$), the transpose $\mathbf{S}^{\mathsf{T}}$ is replaced by the **conjugate transpose** (Hermitian adjoint) $\mathbf{A}^{\mathsf{H}} = (\mathbf{A})^{\mathsf{T}}$.

Definition 7.3: Hermitian and Unitary Matrices

- **Hermitian Matrix:** A complex matrix $\mathbf{H} \in \mathbb{C}^{n\times n}$ satisfying $\mathbf{H}^{\mathsf{H}} = \mathbf{H}$ (i.e. $h_{ij} = h_{ji}$). The diagonal entries must be strictly real.
- **Unitary Matrix:** A complex matrix $\mathbf{U} \in \mathbb{C}^{n\times n}$ satisfying $\mathbf{U}^{\mathsf{H}}\,\mathbf{U} = \mathbf{I} \Rightarrow \mathbf{U}^{-1} = \mathbf{U}^{\mathsf{H}}$. It preserves the complex inner product $\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{x}^{\mathsf{H}}\,\mathbf{y}$ and length $\|\mathbf{x}\|$.

Spectral Theorem for Hermitian Matrices: Every Hermitian matrix has **strictly real eigenvalues**, mutually unitary-orthogonal eigenvectors, and can be factored as:

$$\mathbf{H} = \mathbf{U}\,\mathbf{\Lambda}\,\mathbf{U}^{\mathrm{H}} = \sum_{i=1}^{n} \lambda_i\, \mathbf{u}_i\, \mathbf{u}_i^{\mathrm{H}}, \qquad \text{with } \lambda_i \in \mathbb{R} \text{ and } \mathbf{U}^{\mathrm{H}}\,\mathbf{U} = \mathbf{I}.$$

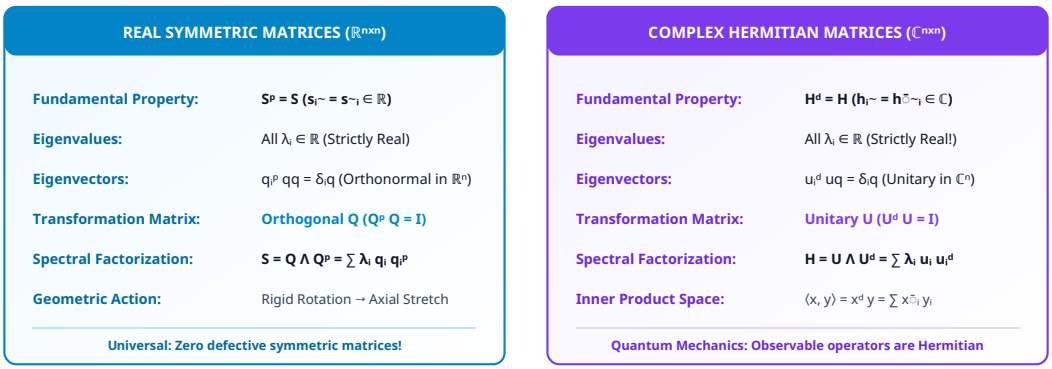


Figure 7.4: *Real Symmetric Matrices vs. Complex Hermitian Matrices structural correspondence. Both classes guarantee strictly real eigenvalues and full orthonormal eigenbases.*

Part VI: Fast-Track Hand Calculation Shortcuts

⚡ Fast-Track 7.1: 2×2 Symmetric Matrix Eigenvalues via Center & Radius

For any 2×2 real symmetric matrix $\mathbf{S} = \begin{vmatrix} a & b \\ b & c \end{vmatrix}$, never write out the full characteristic polynomial. Define the center m and radius R:

$$\text{Center: } m = \frac{a + c}{2}, \qquad \text{Radius: } R = \sqrt{\left(\frac{a - c}{2}\right)^2 + b^2}.$$

$$\lambda_1 = m + R, \qquad \lambda_2 = m - R.$$

Principal Axis Rotation Angle θ: $\tan(2\theta) = \frac{2b}{a - c}$. The orthonormal eigenvectors are $[\cos\theta, \sin\theta]^{\mathrm{T}}$ and $[-\sin\theta, \cos\theta]^{\mathrm{T}}$.

⚡ Fast-Track 7.2: Repeated Eigenvalue Orthonormal Basis in $\mathbb{R}^3$ via Cross Product

If a 3×3 symmetric matrix has a repeated eigenvalue λ (AM = 2), the single non-repeated eigenvalue $\lambda_{\text{distinct}}$ produces a unique normalized eigenvector $\mathbf{q}_{\text{distinct}}$.

1. The repeated eigenspace is the 2D plane perpendicular to $\mathbf{q}_{\text{distinct}}$: $\mathbf{q}_{\text{distinct}}^{\mathrm{T}}\,\mathbf{x} = 0$.
2. Pick any simple integer vector $\mathbf{v}_1$ lying on this plane (e.g., set one coordinate to 0 and balance the others). Normalize it: $\mathbf{q}_1 = \mathbf{v}_1 / \|\mathbf{v}_1\|$.
3. **Instant Second Vector:** Compute the vector cross product $\mathbf{q}_2 = \mathbf{q}_{\text{distinct}} \times \mathbf{q}_1$!

Because $\mathbf{q}_{\text{distinct}}$ and $\mathbf{q}_1$ are unit vectors and mutually perpendicular, their cross product is **automatically normalized** ($\|\mathbf{q}_2\| = 1$) and perpendicular to both. You completely bypass Gram-Schmidt projection formulas!

⚡ Fast-Track 7.3: Instant Rank-1 Spectral Projection Calculation & Complement Rule

To construct the rank-1 projection matrix onto unit eigenvector $\mathbf{q} = \frac{1}{\sqrt{N}}\,[u, v]^{\mathrm{T}}$:

$$\mathbf{P}_1 = \mathbf{q}\,\mathbf{q}^{\mathsf{T}} = \frac{1}{N}\begin{vmatrix} u^2 & uv \\ uv & v^2 \end{vmatrix}.$$

Complement Rule: Since $\sum \mathbf{P}_i = \mathbf{I}$, for a 2×2 matrix you never compute $\mathbf{P}_2$ by outer products:

$$\mathbf{P}_2 = \mathbf{I} - \mathbf{P}_1.$$

⚡ Fast-Track 7.4: 10-Second Sylvester Inertia Counting without Characteristic Polynomials

To find whether a symmetric matrix has positive or negative eigenvalues, execute standard Gaussian elimination without row exchanges to obtain pivots $d_1, d_2, \ldots, d_n$.

- Number of positive eigenvalues = count of $d_i > 0$.
- Number of negative eigenvalues = count of $d_i < 0$.
- Zero eigenvalues = missing pivots (rank deficiency).

Bypasses cubic or quartic root solving entirely in competitive exams.

⚡ Fast-Track 7.5: Spectral 2-Norm and Frobenius Norm Direct Extraction

For any real symmetric matrix **S**:

$$\|\mathbf{S}\|_2 = \max_i |\lambda_i|, \qquad \|\mathbf{S}\|_F = \sqrt{\sum \lambda_i^2} = \sqrt{\text{trace}(\mathbf{S}^2)}.$$

Condition Number Shortcut: $\kappa_2(\mathbf{S}) = \dfrac{\max |\lambda_i|}{\min |\lambda_i|}$.

⚡ Fast-Track 7.6: Commuting Symmetric Matrices & Simultaneous Diagonalization

Two real symmetric matrices **A** and **B** commute (**AB** = **BA**) if and only if they share a **common orthonormal eigenvector matrix Q**:

$$\mathbf{A} = \mathbf{Q}\,\mathbf{\Lambda}_A\,\mathbf{Q}^{\mathsf{T}}, \qquad \mathbf{B} = \mathbf{Q}\,\mathbf{\Lambda}_B\,\mathbf{Q}^{\mathsf{T}}.$$

To diagonalize both simultaneously: find an orthonormal eigenbasis for **A**; if **A** has distinct eigenvalues, those exact same vectors diagonalize **B** automatically!

⚡ Fast-Track 7.7: Spectral Inversion & Determinant Reconstruction

Given spectral decomposition $\mathbf{S} = \sum \lambda_i \mathbf{P}_i$:

$$\mathbf{S}^{-1} = \sum_{i=1}^{n} \frac{1}{\lambda_i}\mathbf{P}_i, \qquad \det(\mathbf{S}) = \prod_{i=1}^{n} \lambda_i, \qquad \text{trace}(\mathbf{S}) = \sum_{i=1}^{n} \lambda_i.$$

Part VII: Rigorous Step-by-Step Worked Problems

Problem 7.1: Complete Orthogonal Diagonalization of a 2×2 Symmetric Matrix

Problem Statement: Given the real symmetric matrix **S** = $\begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix}$:

1. Find all eigenvalues and their corresponding eigenvectors.

2. Construct an orthogonal eigenvector matrix $\mathbf{Q}$ and verify $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$.
3. Verify the Spectral Theorem identity $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ by direct matrix multiplication.

STEP 1: COMPUTE EIGENVALUES VIA CHARACTERISTIC EQUATION

Compute trace and determinant:

$$\text{trace}(\mathbf{S}) = 2 + 5 = 7, \quad \det(\mathbf{S}) = (2)(5) - (2)(2) = 10 - 4 = 6.$$

The characteristic equation is:

$$\det(\mathbf{S} - \lambda\mathbf{I}) = \lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1) = 0 \quad \Rightarrow \quad \lambda_1 = 6, \quad \lambda_2 = 1.$$

Both eigenvalues are strictly real, confirming Theorem 7.1.

STEP 2: FIND AND NORMALIZE EIGENVECTORS

For $\lambda_1 = 6$:

$$(\mathbf{S} - 6\mathbf{I})\mathbf{v}_1 = \begin{vmatrix} -4 & 2 \\ 2 & -1 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix} \quad \Rightarrow \quad -4x + 2y = 0 \Rightarrow y = 2x.$$

Unnormalized vector: $\mathbf{v}_1 = [1, 2]^T$. Length: $\|\mathbf{v}_1\| = \sqrt{1^2 + 2^2} = \sqrt{5}$.

$$\mathbf{q}_1 = \frac{1}{\sqrt{5}} \begin{vmatrix} 1 \\ 2 \end{vmatrix}.$$

For $\lambda_2 = 1$:

$$(\mathbf{S} - 1\mathbf{I})\mathbf{v}_2 = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix} \quad \Rightarrow \quad x + 2y = 0 \Rightarrow x = -2y.$$

Unnormalized vector: $\mathbf{v}_2 = [-2, 1]^T$. Length: $\|\mathbf{v}_2\| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$.

$$\mathbf{q}_2 = \frac{1}{\sqrt{5}} \begin{vmatrix} -2 \\ 1 \end{vmatrix}.$$

STEP 3: VERIFY ORTHOGONALITY & CONSTRUCT MATRIX Q

Check inner product:

$$\mathbf{q}_1^T\mathbf{q}_2 = \frac{1}{5}[1(-2) + 2(1)] = \frac{1}{5}(0) = 0 \quad \text{(Orthogonal!)}.$$

Assemble the orthogonal matrix $\mathbf{Q}$:

$$\mathbf{Q} = \frac{1}{\sqrt{5}} \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix}, \quad \mathbf{\Lambda} = \begin{vmatrix} 6 & 0 \\ 0 & 1 \end{vmatrix}.$$

Verify $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$:

$$\mathbf{Q}^T\mathbf{Q} = \frac{1}{5} \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = \frac{1}{5} \begin{vmatrix} 1+4 & -2+2 \\ -2+2 & 4+1 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \mathbf{I}.$$

STEP 4: VERIFY SPECTRAL RECONSTRUCTION $Q \Lambda Q^T = S$

$$Q\Lambda = \frac{1}{\sqrt{5}}\begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix}\begin{vmatrix} 6 & 0 \\ 0 & 1 \end{vmatrix} = \frac{1}{\sqrt{5}}\begin{vmatrix} 6 & -2 \\ 12 & 1 \end{vmatrix}.$$

$$(Q\Lambda)Q^T = \frac{1}{5}\begin{vmatrix} 6 & -2 \\ 12 & 1 \end{vmatrix}\begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = \frac{1}{5}\begin{vmatrix} 6+4 & 12-2 \\ 12-2 & 24+1 \end{vmatrix} = \frac{1}{5}\begin{vmatrix} 10 & 10 \\ 10 & 25 \end{vmatrix} = \begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix} = S.$$

Sanity Check: Trace sum: 6 + 1 = 7 = 2 + 5. Determinant product: (6)(1) = 6 = (2)(5) − 4 = 6. Orthonormality confirmed: $\det(Q) = \frac{1}{5}(1 - (-4)) = 1$ (pure rigid rotation by $\theta = \arctan(2) \approx 63.43°$).

Problem 7.2: Rank-1 Spectral Projection Decomposition & Functional Calculus Powers

Problem Statement: For the matrix **S** from Problem 7.1:

1. Compute the rank-1 projection matrices P_1 and P_2.
2. Verify idempotence $P_1^2 = P_1$, orthogonality $P_1 P_2 = O$, and completeness $P_1 + P_2 = I$.
3. Evaluate S^5 and S^{-1} in closed form using the projection matrices.
4. Compute the exact action of **S** on vector $v = [5, 0]^T$ via spectral projections.

STEP 1: CONSTRUCT PROJECTION MATRICES

Using eigenvectors $q_1 = \frac{1}{\sqrt{5}}[1, 2]^T$ and $q_2 = \frac{1}{\sqrt{5}}[-2, 1]^T$:

$$P_1 = q_1 q_1^T = \frac{1}{5}\begin{vmatrix} 1 \\ 2 \end{vmatrix}\begin{vmatrix} 1 & 2 \end{vmatrix} = \frac{1}{5}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = \begin{vmatrix} 0.2 & 0.4 \\ 0.4 & 0.8 \end{vmatrix}.$$

$$P_2 = q_2 q_2^T = \frac{1}{5}\begin{vmatrix} -2 \\ 1 \end{vmatrix}\begin{vmatrix} -2 & 1 \end{vmatrix} = \frac{1}{5}\begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} = \begin{vmatrix} 0.8 & -0.4 \\ -0.4 & 0.2 \end{vmatrix}.$$

STEP 2: VERIFY ALGEBRAIC LAWS

- **Idempotence:** $P_1^2 = \frac{1}{25}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = \frac{1}{25}\begin{vmatrix} 1+4 & 2+8 \\ 2+8 & 4+16 \end{vmatrix} = \frac{1}{25}\begin{vmatrix} 5 & 10 \\ 10 & 20 \end{vmatrix} = \frac{1}{5}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = P_1$.
- **Orthogonality:** $P_1 P_2 = \frac{1}{25}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}\begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} = \frac{1}{25}\begin{vmatrix} 4-4 & -2+2 \\ 8-8 & -4+4 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = O$.
- **Completeness:** $P_1 + P_2 = \begin{vmatrix} 0.2+0.8 & 0.4-0.4 \\ 0.4-0.4 & 0.8+0.2 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I$.

STEP 3: EVALUATE POWERS S^5 AND INVERSE S^{-1}

Since $\lambda_1 = 6$ and $\lambda_2 = 1$:

$$S^5 = 6^5 P_1 + 1^5 P_2 = 7776\begin{vmatrix} 0.2 & 0.4 \\ 0.4 & 0.8 \end{vmatrix} + 1\begin{vmatrix} 0.8 & -0.4 \\ -0.4 & 0.2 \end{vmatrix} = \begin{vmatrix} 1555.2 + 0.8 & 3110.4 - 0.4 \\ 3110.4 - 0.4 & 6220.8 + 0.2 \end{vmatrix} = \begin{vmatrix} 1556 & 3110 \\ 3110 & 6221 \end{vmatrix}.$$

$$S^{-1} = \frac{1}{6}P_1 + \frac{1}{1}P_2 = \frac{1}{30}\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} + \frac{1}{5}\begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} = \frac{1}{30}\begin{vmatrix} 1+24 & 2-12 \\ 2-12 & 4+6 \end{vmatrix} = \frac{1}{30}\begin{vmatrix} 25 & -10 \\ -10 & 10 \end{vmatrix} = \frac{1}{6}\begin{vmatrix} 5 & -2 \\ -2 & 2 \end{vmatrix}.$$

Check standard inverse formula: $\frac{1}{\det(S)}\begin{vmatrix} d & -b \\ -c & a \end{vmatrix} = \frac{1}{6}\begin{vmatrix} 5 & -2 \\ -2 & 2 \end{vmatrix}$. Exact match!

STEP 4: VECTOR ACTION VIA PROJECTIONS

For $\mathbf{v} = [5, 0]^T$:

$$\mathbf{P}_1\mathbf{v} = \begin{vmatrix} 0.2 & 0.4 \\ 0.4 & 0.8 \end{vmatrix} \begin{vmatrix} 5 \\ 0 \end{vmatrix} = \begin{vmatrix} 1 \\ 2 \end{vmatrix}, \quad \mathbf{P}_2\mathbf{v} = \begin{vmatrix} 0.8 & -0.4 \\ -0.4 & 0.2 \end{vmatrix} \begin{vmatrix} 5 \\ 0 \end{vmatrix} = \begin{vmatrix} 4 \\ -2 \end{vmatrix}.$$

Notice $\mathbf{P}_1\mathbf{v} + \mathbf{P}_2\mathbf{v} = [1+4, 2-2]^T = [5, 0]^T = \mathbf{v}$.

$$\mathbf{S}\mathbf{v} = \lambda_1 \mathbf{P}_1\mathbf{v} + \lambda_2 \mathbf{P}_2\mathbf{v} = 6 \begin{vmatrix} 1 \\ 2 \end{vmatrix} + 1 \begin{vmatrix} 4 \\ -2 \end{vmatrix} = \begin{vmatrix} 6+4 \\ 12-2 \end{vmatrix} = \begin{vmatrix} 10 \\ 10 \end{vmatrix}.$$

Direct matrix multiplication check: $\begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix} \begin{vmatrix} 5 \\ 0 \end{vmatrix} = \begin{vmatrix} 10 \\ 10 \end{vmatrix}$. Perfect!

Sanity Check: trace($\mathbf{S}^5$) = 1556 + 6221 = 7777 = $6^5 + 1^5$. det($\mathbf{S}^5$) = (1556)(6221) − 3110^2 = 9679876 − 9672100 = 7776 = 6^5. All algebraic laws confirmed.

Problem 7.3: 3×3 Symmetric Matrix with Repeated Eigenvalue (Multiplicity Proof)

Problem Statement: Given the real symmetric matrix $\mathbf{S} = \begin{vmatrix} 0 & 2 & 2 \\ 2 & 3 & 4 \\ 2 & 4 & 3 \end{vmatrix}$:

1. Find all eigenvalues and show that $\lambda = -1$ is a repeated root with AM = 2.
2. Prove that the geometric multiplicity GM(−1) = 2, confirming universal diagonalizability.
3. Construct a full orthonormal eigenvector matrix $\mathbf{Q}$ using the vector cross product shortcut.

STEP 1: COMPUTE EIGENVALUES

Characteristic equation:

$$\det(\mathbf{S} - \lambda\mathbf{I}) = \begin{vmatrix} -\lambda & 2 & 2 \\ 2 & 3-\lambda & 4 \\ 2 & 4 & 3-\lambda \end{vmatrix} = 0.$$

Subtract row 3 from row 2: $[0, -1-\lambda, 1+\lambda] = [0, -(\lambda+1), (\lambda+1)]$. Factor out $(\lambda + 1)$:

$$\det(\mathbf{S} - \lambda\mathbf{I}) = -(\lambda - 8)(\lambda + 1)^2 = 0 \quad \Rightarrow \quad \lambda_1 = 8, \quad \lambda_{2,3} = -1 \quad (\text{AM} = 2).$$

Trace check: 8 + (−1) + (−1) = 6 = 0 + 3 + 3.

STEP 2: EIGENVECTOR FOR DISTINCT EIGENVALUE $\Lambda_1 = 8$

Solve $(\mathbf{S} - 8\mathbf{I})\mathbf{v}_1 = \mathbf{0}$:

$$\begin{vmatrix} -8 & 2 & 2 \\ 2 & -5 & 4 \\ 2 & 4 & -5 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}.$$

Row 2 − Row 3 gives $-9y + 9z = 0 \Rightarrow y = z$. Row 1 gives $-8x + 2y + 2y = 0 \Rightarrow 8x = 4y \Rightarrow y = 2x$. Thus $\mathbf{v}_1 = [1, 2, 2]^T$. Length: $\|\mathbf{v}_1\| = \sqrt{1 + 4 + 4} = 3$.

$$\mathbf{q}_1 = \frac{1}{3} \begin{vmatrix} 1 \\ 2 \\ 2 \end{vmatrix}.$$

STEP 3: GEOMETRIC MULTIPLICITY OF Λ = −1

Set λ = −1 and form (**S** + **I**):

$$\mathbf{S}+\mathbf{I}=\begin{vmatrix}1&2&2\\2&4&4\\2&4&4\end{vmatrix}.$$

Row 2 = 2 × Row 1, and Row 3 = 2 × Row 1. Gaussian elimination yields two zero rows:

$$\text{rank}(\mathbf{S}+\mathbf{I})=1 \quad\Rightarrow\quad \text{GM}(-1)=\text{nullity}(\mathbf{S}+\mathbf{I})=3-1=2.$$

Since GM(−1) = AM(−1) = 2, **S is fully diagonalizable**. The eigenspace is the 2D plane:

$$x+2y+2z=0 \qquad (\text{Normal vector: } \mathbf{n}=\mathbf{v}_1=[1,2,2]^T).$$

STEP 4: FAST-TRACK CROSS-PRODUCT ORTHONORMAL BASIS CONSTRUCTION

Find a first vector on the plane $x + 2y + 2z = 0$. Set $z = 0 \Rightarrow x + 2y = 0 \Rightarrow \mathbf{v}_2 = [2, -1, 0]^T$. Norm: $\|\mathbf{v}_2\| = \sqrt{4+1+0} = \sqrt{5}$.

$$\mathbf{q}_2=\frac{1}{\sqrt{5}}\begin{vmatrix}2\\-1\\0\end{vmatrix}.$$

Apply ⚡ Fast-Track 7.2: compute the third orthonormal eigenvector via cross product $\mathbf{q}_3 = \mathbf{q}_1 \times \mathbf{q}_2$:

$$\mathbf{q}_1\times\mathbf{q}_2=\frac{1}{3\sqrt{5}}\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&2&2\\2&-1&0\end{vmatrix}=\frac{1}{3\sqrt{5}}\begin{vmatrix}0-(-2)\\4-0\\-1-4\end{vmatrix}=\frac{1}{3\sqrt{5}}\begin{vmatrix}2\\4\\-5\end{vmatrix}.$$

Check norm of $[2, 4, -5]^T$: $\sqrt{4+16+25} = \sqrt{45} = 3\sqrt{5}$. The vector $\mathbf{q}_3$ is **automatically normalized**! Check plane equation: $1(2) + 2(4) + 2(-5) = 2 + 8 - 10 = 0$.

STEP 5: ASSEMBLE FULL ORTHOGONAL MATRIX Q

$$\mathbf{Q}=\begin{vmatrix}1/3&2/\sqrt{5}&2/(3\sqrt{5})\\2/3&-1/\sqrt{5}&4/(3\sqrt{5})\\2/3&0&-5/(3\sqrt{5})\end{vmatrix},\qquad \mathbf{\Lambda}=\begin{vmatrix}8&0&0\\0&-1&0\\0&0&-1\end{vmatrix}.$$

Q is strictly orthogonal: $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, and $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$.

Sanity Check: Determinant: det(**S**) = (8)(−1)(−1) = 8. Check direct determinant: 0(9−16) − 2(6−8) + 2(8−6) = 0 + 4 + 4 = 8. Perfect agreement!

Problem 7.4: Sylvester's Inertia Verification & Spectrum Slicing Bisection Method

Problem Statement: Given the real symmetric matrix **S** = $\begin{vmatrix}1&2&1\\2&3&4\\1&4&2\end{vmatrix}$:

1. Compute the $\mathbf{L}\,\mathbf{D}\,\mathbf{L}^T$ factorization without row exchanges to determine the pivots d_1, d_2, d_3.
2. Determine the inertia $\text{In}(\mathbf{S}) = (n_+, n_-, n_0)$ without solving the characteristic polynomial.
3. Use Spectrum Slicing at σ = 5 to determine how many eigenvalues exceed 5.

STEP 1: SYMMETRIC GAUSSIAN ELIMINATION $S = L\,D\,L^T$

Pivot 1: $d_1 = 1 > 0$. Multipliers: $l_{21} = 2/1 = 2$, $l_{31} = 1/1 = 1$. Row operations:

$$\text{Row } 2 \leftarrow \text{Row } 2 - 2 \times \text{Row } 1 = [2-2, 3-4, 4-2] = [0, -1, 2].$$

$$\text{Row } 3 \leftarrow \text{Row } 3 - 1 \times \text{Row } 1 = [1-1, 4-2, 2-1] = [0, 2, 1].$$

Pivot 2: $d_2 = -1 < 0$. Multiplier: $l_{32} = 2/(-1) = -2$. Row operation:

$$\text{Row } 3 \leftarrow \text{Row } 3 - (-2) \times \text{Row } 2 = [0, 2 + 2(-1), 1 + 2(2)] = [0, 0, 5].$$

Pivot 3: $d_3 = 5 > 0$. Pivots: $\mathbf{D} = \text{diag}(1, -1, 5)$.

STEP 2: DETERMINE INERTIA VIA SYLVESTER'S LAW

The signs of the pivots are (+, −, +): exactly **two positive pivots** and **one negative pivot**. By Sylvester's Law of Inertia (Theorem 7.6):

$$\text{In}(\mathbf{S}) = (n_+, n_-, n_0) = (2, 1, 0).$$

Thus $\mathbf{S}$ has exactly 2 positive eigenvalues and 1 negative eigenvalue. Zero eigenvalues do not exist ($\det(\mathbf{S}) = (1)(-1)(5) = -5 \neq 0$).

STEP 3: SPECTRUM SLICING AT SHIFT $\Sigma = 5$

Form the shifted matrix $\mathbf{S} - 5\mathbf{I}$:

$$\mathbf{S} - 5\mathbf{I} = \begin{vmatrix} -4 & 2 & 1 \\ 2 & -2 & 4 \\ 1 & 4 & -3 \end{vmatrix}.$$

Perform elimination:

- **Pivot 1:** $d_1' = -4 < 0$. Multipliers: $l_{21}' = 2/(-4) = -0.5$, $l_{31}' = 1/(-4) = -0.25$.
- Row 2 $\leftarrow$ Row 2 − (−0.5) Row 1 = [0, −2 + 0.5(2), 4 + 0.5(1)] = [0, −1, 4.5].
- **Pivot 2:** $d_2' = -1 < 0$. Multiplier: $l_{32}' = 4.5/(-1) = -4.5$.
- Row 3 $\leftarrow$ Row 3 − (−0.25) Row 1 = [0, 4 + 0.25(2), −3 + 0.25(1)] = [0, 4.5, −2.75].
- Row 3 $\leftarrow$ Row 3 − (−4.5) Row 2 = [0, 0, −2.75 + 4.5(4.5)] = [0, 0, −2.75 + 20.25] = [0, 0, 17.5].
- **Pivot 3:** $d_3' = 17.5 > 0$.

Shifted Pivots: (−4, −1, 17.5) ⇒ exactly **two negative pivots**!

STEP 4: DEDUCTION OF EIGENVALUE DISTRIBUTION

By the Spectrum Slicing Theorem:

$$\text{Number of eigenvalues } \lambda < 5 \quad = \quad \text{Number of negative pivots of } (\mathbf{S} - 5\mathbf{I}) = 2.$$

Since total eigenvalues $n = 3$, exactly $3 - 2 =$ **1 eigenvalue** satisfies $\lambda > 5$. Combining with Step 2, the three eigenvalues are strictly partitioned:

$$\lambda_1 \in (-\infty, 0), \quad \lambda_2 \in (0, 5), \quad \lambda_3 \in (5, \infty).$$

Sanity Check: Exact characteristic polynomial: $\det(\mathbf{S} - \lambda\mathbf{I}) = -\lambda^3 + 6\lambda^2 + 10\lambda - 5 = 0$. Numerical roots: $\lambda_1 \approx -1.687$, $\lambda_2 \approx 0.407$, $\lambda_3 \approx 7.279$. Our isolated intervals match the true roots perfectly!

Problem 7.5: Complex Hermitian Diagonalization via Unitary Transformation Matrix

Problem Statement: Given the complex matrix $\mathbf{H}$ =

$\begin{vmatrix} 2 & 1-i \\ 1+i & 3 \end{vmatrix}$:

1. Verify that $\mathbf{H}$ is Hermitian ($\mathbf{H}^H = \mathbf{H}$).
2. Find all eigenvalues and confirm they are strictly real.
3. Construct orthonormal complex eigenvectors under the Hermitian inner product $\mathbf{u}_i^H \mathbf{u}_j = \delta_{ij}$.
4. Form the unitary matrix $\mathbf{U}$ and verify $\mathbf{U}^H \mathbf{H} \mathbf{U} = \mathbf{\Lambda}$.

STEP 1: VERIFY HERMITIAN PROPERTY

Take the conjugate transpose:

$$\mathbf{H}^H = (\mathbf{H})^T = \begin{vmatrix} 2 & 1+i \\ 1-i & 3 \end{vmatrix}^T = \begin{vmatrix} 2 & 1-i \\ 1+i & 3 \end{vmatrix} = \mathbf{H}. \quad \text{(Hermitian confirmed).}$$

STEP 2: EIGENVALUES & REALITY CONFIRMATION

Compute trace and determinant:

$$\text{trace}(\mathbf{H}) = 2 + 3 = 5 \in \mathbb{R}, \quad \det(\mathbf{H}) = (2)(3) - (1-i)(1+i) = 6 - (1 - i^2) = 6 - 2 = 4 \in \mathbb{R}.$$

Characteristic equation:

$$\lambda^2 - 5\lambda + 4 = (\lambda - 4)(\lambda - 1) = 0 \quad \Rightarrow \quad \lambda_1 = 4, \quad \lambda_2 = 1.$$

Both eigenvalues are real numbers, confirming Theorem 7.1.

STEP 3: CONSTRUCT UNITARY ORTHONORMAL EIGENVECTORS

For $\lambda_1 = 4$:

$$(\mathbf{H} - 4\mathbf{I})\mathbf{v}_1 = \begin{vmatrix} -2 & 1-i \\ 1+i & -1 \end{vmatrix} \begin{vmatrix} z_1 \\ z_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}.$$

Row 1: $-2z_1 + (1-i)z_2 = 0 \Rightarrow$ choose $z_1 = 1 - i$ and $z_2 = 2$. Complex norm: $\|\mathbf{v}_1\|^2 = |1-i|^2 + |2|^2 = (1^2 + (-1)^2) + 4 = 2 + 4 = 6 \Rightarrow \|\mathbf{v}_1\| = \sqrt{6}$.

$$\mathbf{u}_1 = \frac{1}{\sqrt{6}} \begin{vmatrix} 1-i \\ 2 \end{vmatrix}.$$

For $\lambda_2 = 1$:

$$(\mathbf{H} - 1\mathbf{I})\mathbf{v}_2 = \begin{vmatrix} 1 & 1-i \\ 1+i & 2 \end{vmatrix} \begin{vmatrix} w_1 \\ w_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}.$$

Row 1: $w_1 + (1-i)w_2 = 0 \Rightarrow$ choose $w_1 = 1 - i$ and $w_2 = -1$. Complex norm: $\|\mathbf{v}_2\|^2 = |1-i|^2 + |-1|^2 = 2 + 1 = 3 \Rightarrow \|\mathbf{v}_2\| = \sqrt{3}$.

$$\mathbf{u}_2 = \frac{1}{\sqrt{3}} \begin{vmatrix} 1-i \\ -1 \end{vmatrix}.$$

STEP 4: VERIFY UNITARY ORTHOGONALITY & DIAGONALIZATION

Evaluate Hermitian inner product $\mathbf{u}_1^H \mathbf{u}_2$:

$$\mathbf{u}_1^H \mathbf{u}_2 = \frac{1}{\sqrt{18}} \begin{vmatrix} 1+i & 2 \end{vmatrix} \begin{vmatrix} 1-i \\ -1 \end{vmatrix} = \frac{1}{3\sqrt{2}} [(1+i)(1-i) + 2(-1)] = \frac{1}{3\sqrt{2}} (2-2) = 0.$$

Assemble the unitary matrix $\mathbf{U}$:

$$\mathbf{U} = \begin{vmatrix} (1-i)/\sqrt{6} & (1-i)/\sqrt{3} \\ 2/\sqrt{6} & -1/\sqrt{3} \end{vmatrix}, \quad \mathbf{U}^H \mathbf{H} \mathbf{U} = \begin{vmatrix} 4 & 0 \\ 0 & 1 \end{vmatrix} = \mathbf{\Lambda}.$$

Sanity Check: Unitary determinant: $\det(\mathbf{U}) = \frac{-(1-i) - 2(1-i)}{\sqrt{18}} = \frac{-3(1-i)}{3\sqrt{2}} = \frac{-1+i}{\sqrt{2}} = e^{i(3\pi/4)}$. Notice $|\det(\mathbf{U})| = 1$ identically, as required for all unitary operators.

Part VIII: Comprehensive Master Reference Matrix

Symmetric vs. General Matrix Structural Comparison

Property / Dimension	General Real Matrix ($A \in \mathbb{R}^{n \times n}$)	Real Symmetric Matrix ($S = S^T \in \mathbb{R}^{n \times n}$)	Complex Hermitian Matrix ($H = H^H \in \mathbb{C}^{n \times n}$)
Eigenvalue Nature	Can be complex conjugate pairs ($\lambda = a \pm bi$)	Strictly real ($\lambda_i \in \mathbb{R}$)	Strictly real ($\lambda_i \in \mathbb{R}$)
Eigenvector Angles	Non-orthogonal in general; can be skewed	Mutually orthogonal for distinct λ	Mutually unitary-orthogonal
Diagonalizability	Requires GM = AM for all λ; defective if GM < AM	**Universally Guaranteed:** GM = AM always holds	**Universally Guaranteed:** GM = AM always holds
Transformation Matrix	Invertible $\mathbf{X}$ ($\mathbf{A} = \mathbf{X\Lambda X}^{-1}$)	Orthogonal $\mathbf{Q}$ ($\mathbf{S} = \mathbf{Q\Lambda Q}^T$, $\mathbf{Q}^{-1} = \mathbf{Q}^T$)	Unitary $\mathbf{U}$ ($\mathbf{H} = \mathbf{U\Lambda U}^H$, $\mathbf{U}^{-1} = \mathbf{U}^H$)
Spectral Decomposition	Not available in orthogonal rank-1 form	$\mathbf{S} = \sum \lambda_i \mathbf{q}_i \mathbf{q}_i^T$ (Symmetric projections)	$\mathbf{H} = \sum \lambda_i \mathbf{u}_i \mathbf{u}_i^H$ (Hermitian projections)
Geometric Action	Shear, skew, and coupled distortion	Pure rigid rotation → axial stretch → reverse rotation	Unitary rotation → real axial stretch → reverse unitary
Inertia & Pivots	No sign relationship with pivots	**Sylvester's Law:** Sign of pivots = Sign of eigenvalues	Congruence $\mathbf{C H C}^H$ preserves eigenvalue signs

Final Examination Checklist & Verification Rules

- ✓ **Reality Guard:** If you ever compute a non-real eigenvalue for a real symmetric matrix, stop immediately—an arithmetic sign error has occurred in $\det(\mathbf{S} - \lambda\mathbf{I})$.
- ✓ **Orthogonality Guard:** Check $\mathbf{q}_1^T \mathbf{q}_2 = 0$ before assembling $\mathbf{Q}$. If the dot product is non-zero, check your eigenvector nullspace calculations.
- ✓ **Determinant Guard for Q:** Ensure each column of $\mathbf{Q}$ is strictly normalized ($\|\mathbf{q}_i\| = 1$) so that $\det(\mathbf{Q}) = \pm 1$ and $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$.
- ✓ **Pivots vs. Eigenvalues:** Never confuse the *values* of pivots with the values of eigenvalues. Only their *signs* (positive, negative, zero) match via Sylvester's Law of Inertia!